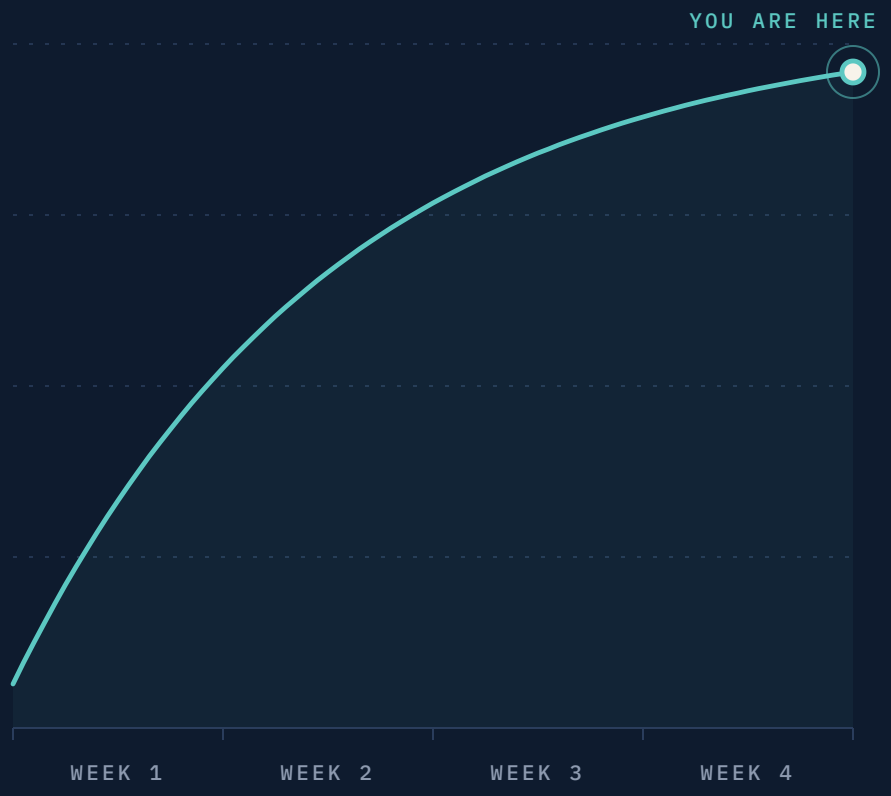
WEEK 4 · DAY 5 · OVERVIEW

Course Project



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ABOUT 40 MIN / 10 SECTIONS

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Brief

Operations sent October's extract with a cover note. Before any number leaves the desk, tell me whether it can be trusted: fix what you can prove, in one function, so November is one call. Then tell me what is in the book, where it is concentrated, and how it differs from the benchmark. Send me a workbook and one page. Describe it: no views on any bond, issuer, or sector.

ABOUT 40 MIN / 10 SECTIONS

The Project on One Page

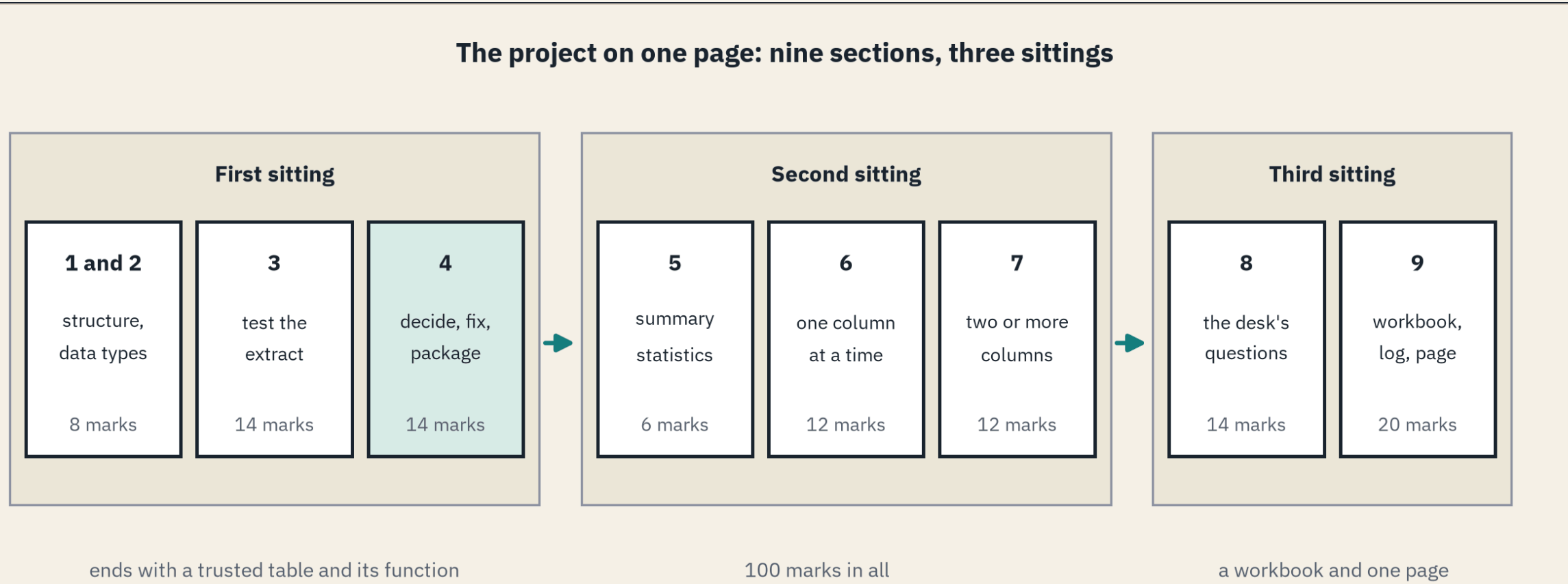


FIGURE 1 The project's nine marked sections in three sittings, each with its marks.

Your Route Through Day 5

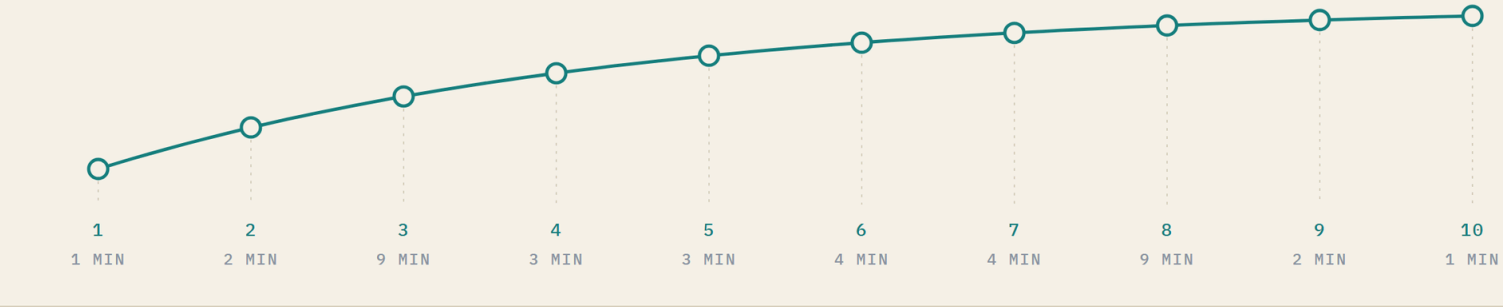


FIGURE 3 Ten sections, in order, with about how long each takes.

	SECTION	ABOUT
1	The Structure of the Data	1 min
2	Data Types	2 min
3	Data Quality: Test the Extract	9 min
4	Data Quality: Decide, Fix, and Package	3 min
5	Summary Statistics	3 min

	SECTION	ABOUT
6	One Column at a Time	4 min
7	Two or More Columns	4 min
8	The Desk's Questions	9 min
9	The Deliverables	2 min
10	Before You Hand In	1 min

Time. About 38 minutes for the lesson.

1 The Structure of the Data

One row of the extract: every column, its value, and its unit

as_of_date	2026-10-30	the date the book is valued
cusip	99080CAE8	the bond's identifier, nine characters
ticker	BEZE	the issuer's ticker
issuer	Bezane Finance	the issuer's name, invented
sector	Financials	a name from the benchmark's list
rating	BBB-	the bond's rating, AAA to CCC
coupon	7.0	annual coupon, percent
maturity	2032-06-18	the date the bond repays
price	106.576	clean price, per 100 of face
price_date	2026-10-30	the date of the price
accrued	2.567	accrued interest, per 100 of face
yield_pct	5.62	yield to maturity, percent
oas	149	option-adjusted spread, basis points
duration	4.545	modified duration, years
spread_duration	4.545	spread duration, years
dts	677.2	spread duration times OAS
amount_outstanding	1,650,000,000	face of the whole issue, dollars
par_held	4,000,000	face the desk holds, dollars
market_value	4,365,720.0	face times (price + accrued) / 100, dollars

4,000,000 x (106.576 + 2.567) / 100 = 4,365,720.00

FIGURE 2 The extract's first row, column by column, and the market value rule: face times price plus accrued, over 100.

THE COVER NOTE'S TOTALS

```
Control totals for the book:
  Number of bonds:      206
  Total face:           507,250,000 dollars
  Total market value:   509,928,477.50 dollars
```

QUESTION	WHAT IT ASKS
1.1 Rows, Columns, and Identifiers	What shape is the extract (**answer_1_1a**), and what is one row? Show the first rows, and count the different as-of dates, CUSIPs (**answer_1_1b**), tickers, and issuers.

Write down the row count: the cover note says 206.

Check that every column holds the right type.

2 Data Types

THE UNITS

Units: **coupon** and **yield_pct** in percent, **oas** in basis points, **duration** in years, **price** and **accrued** per 100 of face, and **par_held**, **amount_outstanding**, and **market_value** in dollars.

THE LOADED FILES

```
raw = pd.read_csv(data_folder + 'project_holdings.csv')
ratings = pd.read_csv(data_folder + 'ratings.csv')
benchmark = pd.read_csv(data_folder + 'benchmark.csv')
```

QUESTION	WHAT IT ASKS
2.1 The Type of Each Column	Which types did pandas give the columns? How many are numbers (**answer_2_1**), and which hold dates?
2.2 A Working Copy With Real Dates	Copy **raw** to **book** and convert its three date columns. How many dates could not be read (**answer_2_2**)? Print each column's earliest and latest.

From here on every fix goes to book**.**

3 Data Quality: Test the Extract

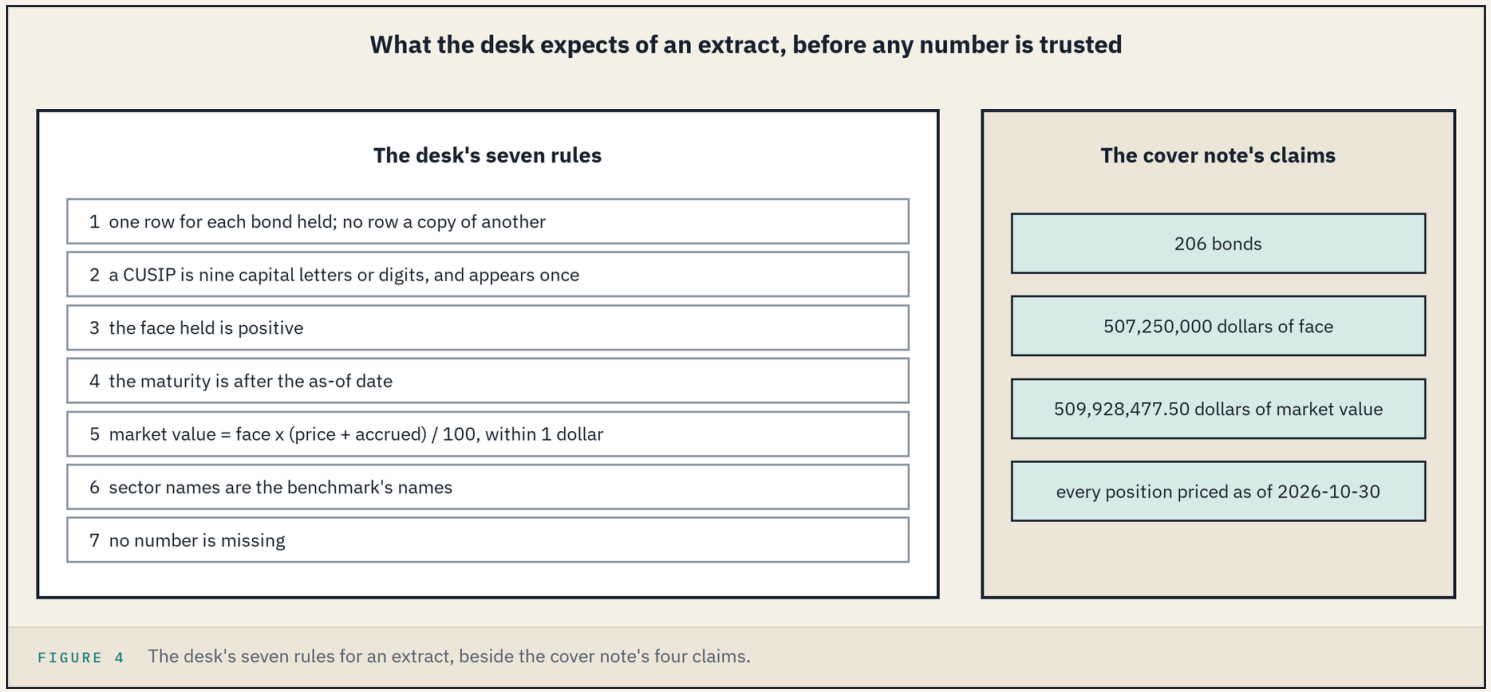


FIGURE 4 The desk's seven rules for an extract, beside the cover note's four claims.

EXCEL SIDE BY SIDE

RULE	EXCEL, ANY CURRENT VERSION
1	`=COUNTIF($B$2:$B$210, B2) > 1` marks rows sharing a CUSIP
2	`=LEN(B2) = 9` tests only the length
3	`=COUNTIF(R2:R210, "<=0")`
5	`=ABS(R2 * (I2 + K2) / 100 - S2) > 1` filled down
6	`=COUNTIF(benchmark!$E$2:$E$822, E2) = 0` ignores capitals, so it misses names `isin()` catches
7	`=COUNTBLANK(M2:M210)`

Every rule gets a count: failures are the fix list, passes go in the page.

4 Data Quality: Decide, Fix, and Package

QUESTION	WHAT IT ASKS
4.1 Decide and Fix	Sort each failing row into the EDA notebook's three kinds: fix, flag, or keep. Give each fix a decision cell, then make it with `headline()` before and after. Rerun 3.2 to 3.9: how many rows still fail (**answer_4_1**)?
Checkpoint: the Cleaned Table	Does **book** match the cover note and the expected decisions? If a line is not `correct`, open that part's Stuck block.
The Three Gaps, Closed	Did every gap to the cover note reach zero? This cell is written for you.
4.2 The Cleaning as a Function	Write `clean_extract(raw)`, its docstring listing 2.2's dates and 4.1's fixes in order, leaving its input unchanged; a second table it needs is a second parameter. Store its result in **cleaned**, `assert` the three totals with messages, and show `cleaned` equals **book**.

THE RULES

- **Work in a copy:** fixes go to **book**, never to `raw`.
- **State every decision** in a markdown cell: what, how many rows, what evidence.
- **Print `headline()` before and after every fix.**

The extract ties to its cover note, and one call rebuilds it. The first sitting ends here.

5 Summary Statistics

QUESTION	WHAT IT ASKS
5.1 The Rating Scale	Join the scale's **score**, **bucket**, and **grade** to **book**, with rows before and after. How many bonds are High Yield (**answer_5_1**)?
5.2 Summary, Mean, Median, and Skew	Summarize **coupon**, **price**, **yield_pct**, **oas**, **duration**, and **market_value**: could a bond have each range? Give the median OAS to 1 place (**answer_5_2a**). Add **dts** and show mean, median, and skew: which column has the longest right tail, and which leans left? Give the OAS skew to 2 places (**answer_5_2b**).
5.3 Counts by Category	Count bonds by sector, by bucket from AAA to CCC, and by grade, drawn as three labelled bar charts in a row. Which bucket is largest (**answer_5_3**)?

THE RATING SCALE

ratings.csv is the rating scale: **rating**, **score** (1 for AAA to 18 for CCC), **bucket**, and **grade**.

THE BENCHMARK

benchmark.csv holds 821 bonds, a **weight_pct** for each, and no **par_held**.

Counts treat every bond alike; section 8 weighs them by value.

6 One Column at a Time

QUESTION	WHAT IT ASKS
6.1 The Distribution of the OAS	Draw a histogram of the OAS with labelled lines at the mean and median. How many bonds sit above the mean (**answer_6_1**)?
6.2 The IQR Rule	Print the OAS quartiles, IQR, and limits, and draw its box plot. Give the upper limit to 3 places (**answer_6_2a**) and the bonds above it (**answer_6_2b**).
6.3 Errors or Real Bonds?	Show the flagged bonds widest first with rating, coupon, price, yield, and OAS. Errors or real? Give the evidence and your decision. Give their share of value, percent to 2 places (**answer_6_3**).
6.4 The Distribution of the Price	Draw a histogram of the clean price with a line at par. How many are below 100 (**answer_6_4**)? Is the lowest or highest priced bond an error?
6.5 The Size of the Positions	Draw a histogram of market value in millions of dollars. What share is the largest position, percent to 2 places (**answer_6_5**)? Does the IQR rule flag any?

Far is not wrong: every flagged bond is tested before it is judged.

7 Two or More Columns

QUESTION	WHAT IT ASKS
7.1 Correlations	Draw the correlations of `coupon`, `price`, `yield_pct`, `oas`, `duration`, and `score` as an annotated heatmap. Name the strongest pair and one near zero. What does OAS against score mean (**answer_7_1**, 2 places)?
7.2 Yield Against Duration, by Grade	Draw yield against duration, coloured by grade. Give their correlation within Investment Grade to 2 places (**answer_7_2**): what does the whole-book figure hide?
7.3 OAS by Rating Bucket	Show each bucket's bond count and median, lowest, and highest OAS in credit order, and draw the medians as bars. Give the BBB median to 1 place (**answer_7_3**).
7.4 OAS by Grade	Draw one OAS box per grade, and print each grade's count and median. Give the High Yield median less the Investment Grade median to 1 place (**answer_7_4**).
7.5 Sector by Grade	Pivot market value in millions of dollars, sectors down and grades across, draw it as a heatmap, and check the cells add to the book. Which sector holds the most High Yield value (**answer_7_5**)?

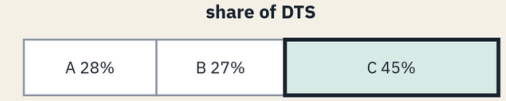
> **A whole-book figure can hide what each grade does.**

8 The Desk's Questions

A bond's share of the book's DTS: three invented bonds, not the file's

bond	weight	DTS	weight x DTS	share of DTS
Bond A	50%	500	0.5 x 500 = 250	28%
Bond B	30%	800	0.3 x 800 = 240	27%
Bond C	20%	2,000	0.2 x 2,000 = 400	45%





book: 890

the contributions add up to the book's DTS

Bond C: 20% of the weight, 45% of the DTS

FIGURE 5 Three invented bonds: the one with 20 percent of the weight carries 45 percent of the book's DTS.

EXCEL SIDE BY SIDE

FIGURE	EXCEL
A sector's weight	`=SUMIF(E2:E207, "Materials", S2:S207) / SUM(S2:S207) * 100`, or a PivotTable value shown as % of Grand Total
Weighted OAS	`=SUMPRODUCT(M2:M207, S2:S207) / SUM(S2:S207)`
The book's DTS	`=SUMPRODUCT(P2:P207, S2:S207) / SUM(S2:S207)`

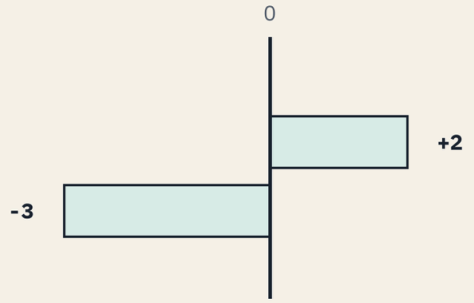
Weights say where the dollars are; DTS shares say where the spread risk is.

Book Less Benchmark

QUESTION	WHAT IT ASKS
8.6 Book Less Benchmark, by Sector	Show each sector's book weight, benchmark weight, and difference in percentage points, and draw the differences as bars with a zero line and a title naming both files' months. Give the largest positive difference's sector (**answer_8_6a**) and size to 2 places (**answer_8_6b**).
8.7 Book Less Benchmark, by Bucket	Join the scale to the benchmark, then compare by bucket in credit order and by grade, with the bucket differences drawn as bars. Give the BBB difference to 2 places (**answer_8_7**).

Book less benchmark: two invented sectors

sector	book	benchmark	difference	called
Sector A	12%	10%	+2 points	overweight
Sector B	5%	8%	-3 points	underweight



Overweight and underweight name the sign of a subtraction, not a view

FIGURE 6 Overweight and underweight name only the sign of the book's share less the benchmark's, not a view.

9 The Deliverables

QUESTION	WHAT IT ASKS
9.1 The Workbook	Write **`project_report.xlsx`**: a **`book`** sheet without row labels and a **`summary`** sheet of 8.1's sector table. Read both back into **answer_9_1a** and **answer_9_1b** as row counts, **`assert`** the book sheet's value, delete the file, and store whether it exists in **answer_9_1c**.
9.2 The AI-Use Log	One row per question you asked an AI assistant; **`none used`** is valid.
9.3 The Page	Write the page for the portfolio manager below, 250 to 350 words, in five parts: the book in five numbers; concentrations by sector, grade, and issuer; data quality, each problem with its rows and its fix, and what was clean; the benchmark, with both files' dates; and open items for operations.

THE AI-USE LOG

WHAT I ASKED	WHAT I KEPT OR REJECTED	HOW I CHECKED IT

Run the notebook clean, then hand it in.

Before You Hand In

WHAT TO HAND IN

- every check **`correct`** and **`raw`** unchanged;
- a decision cell and headlines for every fix;
- **`clean_extract`** with its docstring and passing **`asserts`**;
- no **`.xlsx`** file left behind, and every chart finished;
- the log and the page.

THE RULES, CONTINUED

- **Describe:** never recommend, rank, or forecast.
- **Finish every chart:** title, labelled axes, units.
- **Restart and Run All** before you hand in.
- **Check any AI answer** as week 1's Day 1, Working With an AI Assistant, taught: read, predict, run, check; log it for 9.2.

HOW THE WORK IS READ

The picture's marks show the weight; a number with no sentence earns part of its mark.

The planted problems are in `data/project_answer_key.md`: read it after you finish.

You Can Now

At Day 0 you had never opened a notebook. Since then you have learned to:

- read a traceback;
- filter, group, and join a book;
- draw a finished chart;
- test a file before trusting it;
- pull identifiers from text;
- keep a module, with tests;
- check an AI's code against your own number.

NEXT

WEEK 1 · DAY 1

Time Value of Money

What 100 grows to, what 100 due later is worth today, and your first bond math functions, tested against Excel and saved with Git.